CLM

EXECUTIVE BRIEFING

Fast, cheap, private text classification

When a small trained model beats calling a frontier language model

High-volume classification is a different job

- Millions of short, repetitive decisions: route, tag, flag
- A fixed set of labels, and past examples of each
- Needs to be fast, cheap per decision, private and consistent
- A frontier model is a generalist, paid per token, run remotely

THE APPROACH

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A frozen reader, a small learner, one fast answer

NVIDIA Nemotron reads

Turns each text into numbers. Never retrained.

A small model learns

Our labels, from our examples, in seconds to minutes.

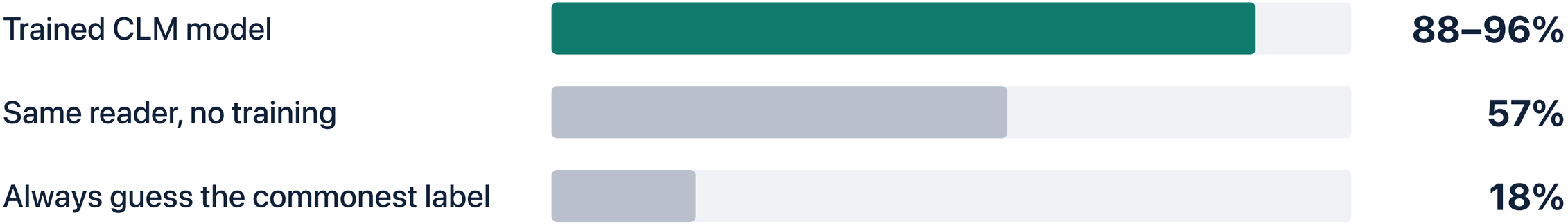
We get the odds

A probability per label, in milliseconds, on our hardware.

ACCURACY (MEASURED)

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Trained on 300 examples: 88% to 96% right



Support-ticket topics, 5 labels, 100 held-out conversations; 88–96% across 9 training runs (mean 92%). Demo data: synthetic, labelled by a model. Frontier accuracy: measure on our data with Marketing/benchmark/compare_frontier.py.

SPEED (MEASURED)

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33 ms

median answer, 43 ms at the 95th percentile, on one laptop

- 36 answers per second per laptop: about 3.2 million a day
- No network hop, no queue at a provider, no rate limits
- Repeated texts answer from cache in about 1 ms

Apple M5 Pro laptop, Nemotron-3-Embed-1B, end to end. A GPU server is faster. Frontier latency includes network and generation time; the comparison script measures it.

COST PER MILLION DECISIONS

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7× to 300× cheaper per million decisions



CLM: 7.6 compute-hours per million at 36/s, priced at an assumed $1.00/hour. Frontier: Anthropic list prices (Oct 2026) at an estimated 330 input and 50 output tokens per ticket. A floor: thinking tokens add to it. The Batch API halves frontier prices but adds hours of delay.

CONTROL

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It stays ours: data, decisions and evidence

- Texts never leave our machines; no per-request third party
- Same input, same answer: training and serving are reproducible
- Verified, versioned model files; a full governance pack in the repo
- Commercial-use encoder from NVIDIA, built on Mistral AI; lineage documented

Use each where it is strongest

	CLM	FRONTIER MODEL
Fixed labels, with examples	**Best fit**	Works, at a cost
Cost per million decisions	**≈ $8**	$58 to $2,320
Answer in milliseconds	**Yes**	Adds network and generation time
Data must stay inside	**Yes**	Leaves our network
Open questions, writing, reasoning	**No**	**Best fit**
No examples, labels change daily	**No**	**Best fit**

Best of both: let CLM decide, and send its low-confidence cases to a frontier model or a person.

Days, not quarters

300+ labelled examples

Exported from the system that already handles the work.

Minutes to train

In a browser: upload, describe the labels, train, test.

One laptop or small server

No GPU required; standalone, no external calls.

Before real decisions: human-checked labels, calibrated probabilities, and the new-model checklist.

RECOMMENDATION

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A four-week pilot on one workflow

- Pick one high-volume routing or triage workflow
- Label 500 real examples; train and test CLM
- Run the same test through a frontier model with the included script
- Decide on measured accuracy, cost and latency, with governance sign-off